

AnchorMoE: Interpretable Time Series Classification via Anchor-Routed MoE

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AnchorMoE unifies classification and explanation: the same patch-to-expert routes predict the class and locate its key patches.

◆ Introduction & Motivation

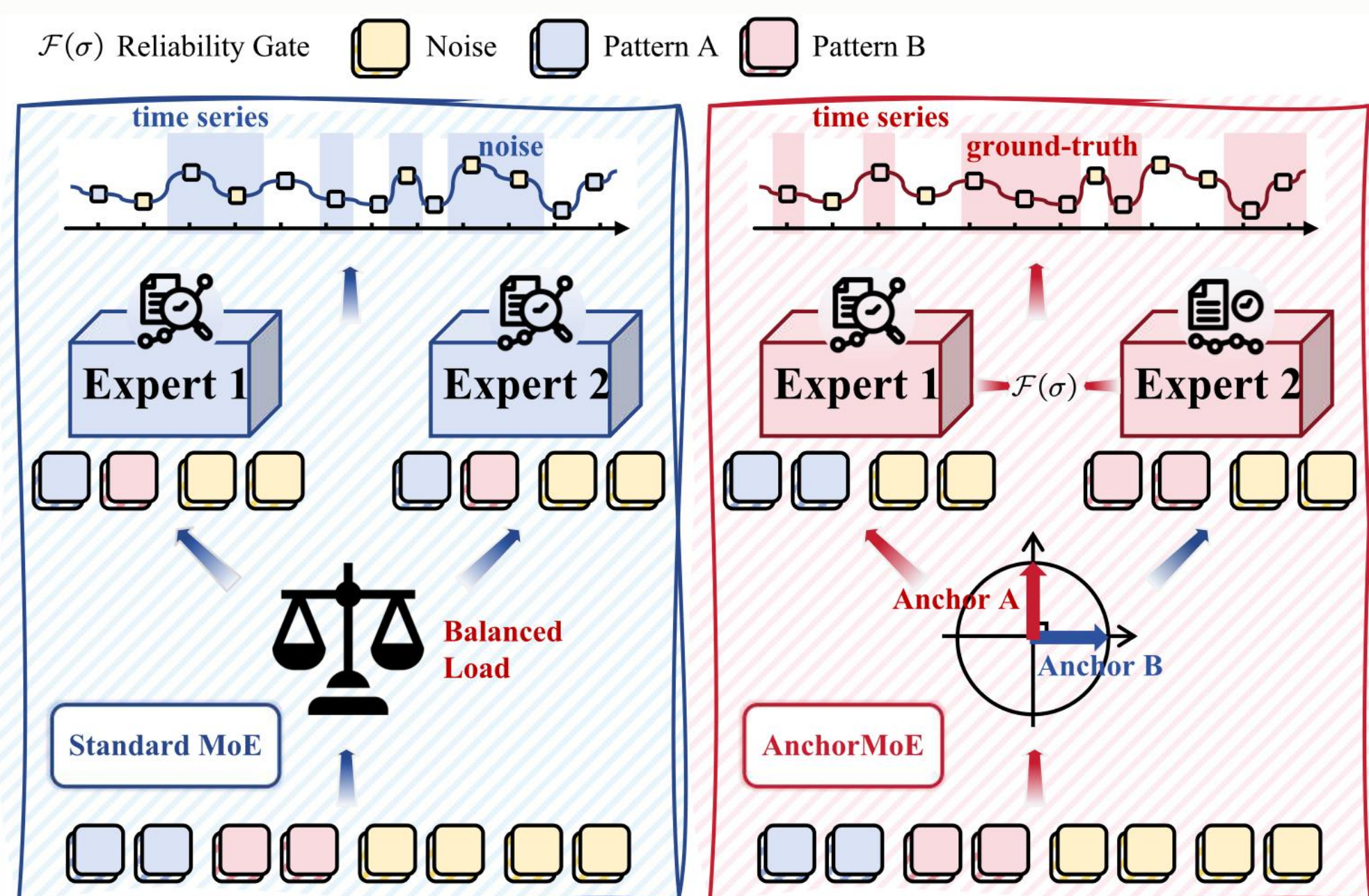


Figure 1. Motivation. Standard MoE learns similar expert roles and absorbs background noise; AnchorMoE separates experts and filters unreliable patches.

◆ AnchorMoE Architecture

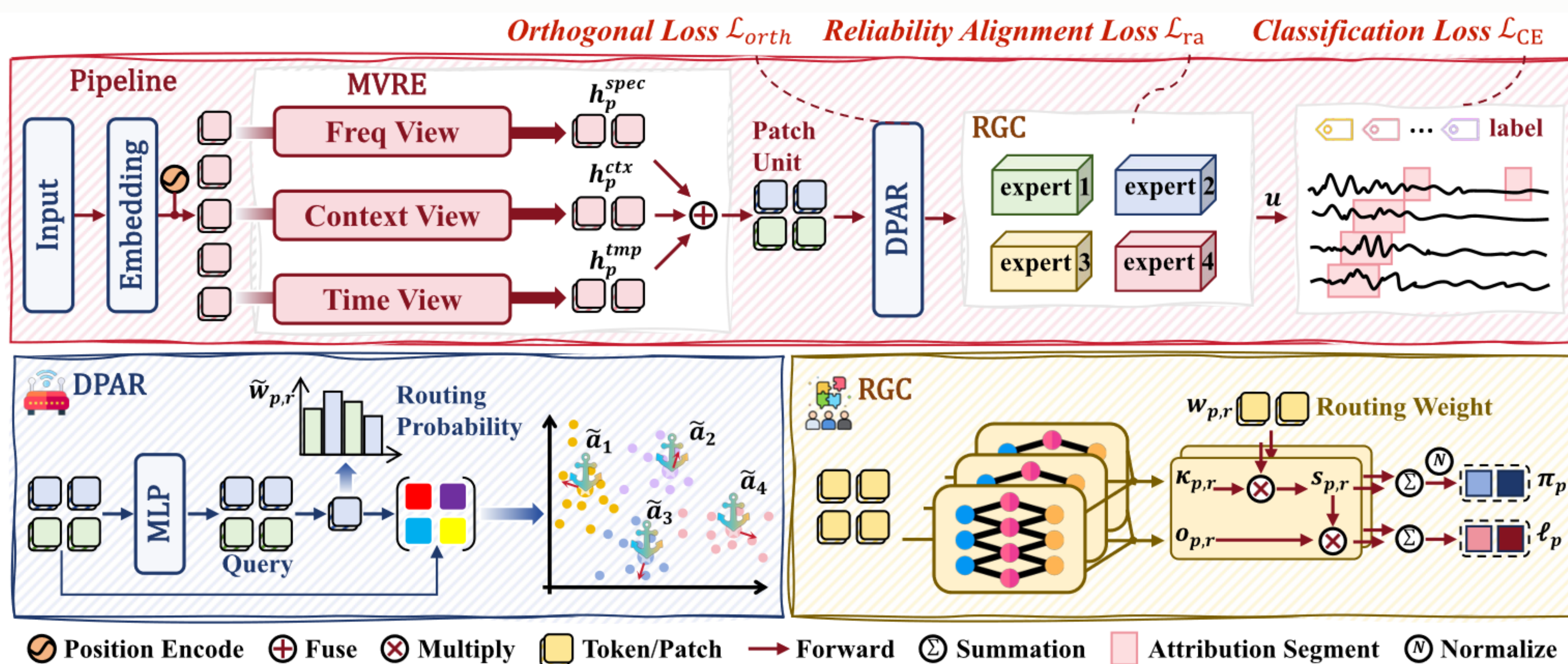


Figure 2. Architecture. MVRE describes each patch, DPAR separates expert roles, and RGC filters unreliable patches before prediction.

One Routed Expert Architecture, Two Outputs

MVRE · Encode Each Patch

Combines time, frequency, and whole-series context to describe each patch.

DPAR · Separate Expert Roles

Separates expert roles so different experts focus on different patch patterns.

RGC · Filter Unreliable Patches

Downweights unreliable experts and background patches before they affect the prediction.

Class Prediction · Key-Patch Explanation

Both come from the same routed patch paths.

$$u_c = \sum_p A_{p,c},$$

$$A_{p,c} = \pi_p \sum_r w_{p,r} \kappa_{p,r} o_{p,r,c}.$$

◆ Experimental Results

Statistical Significance Across Datasets

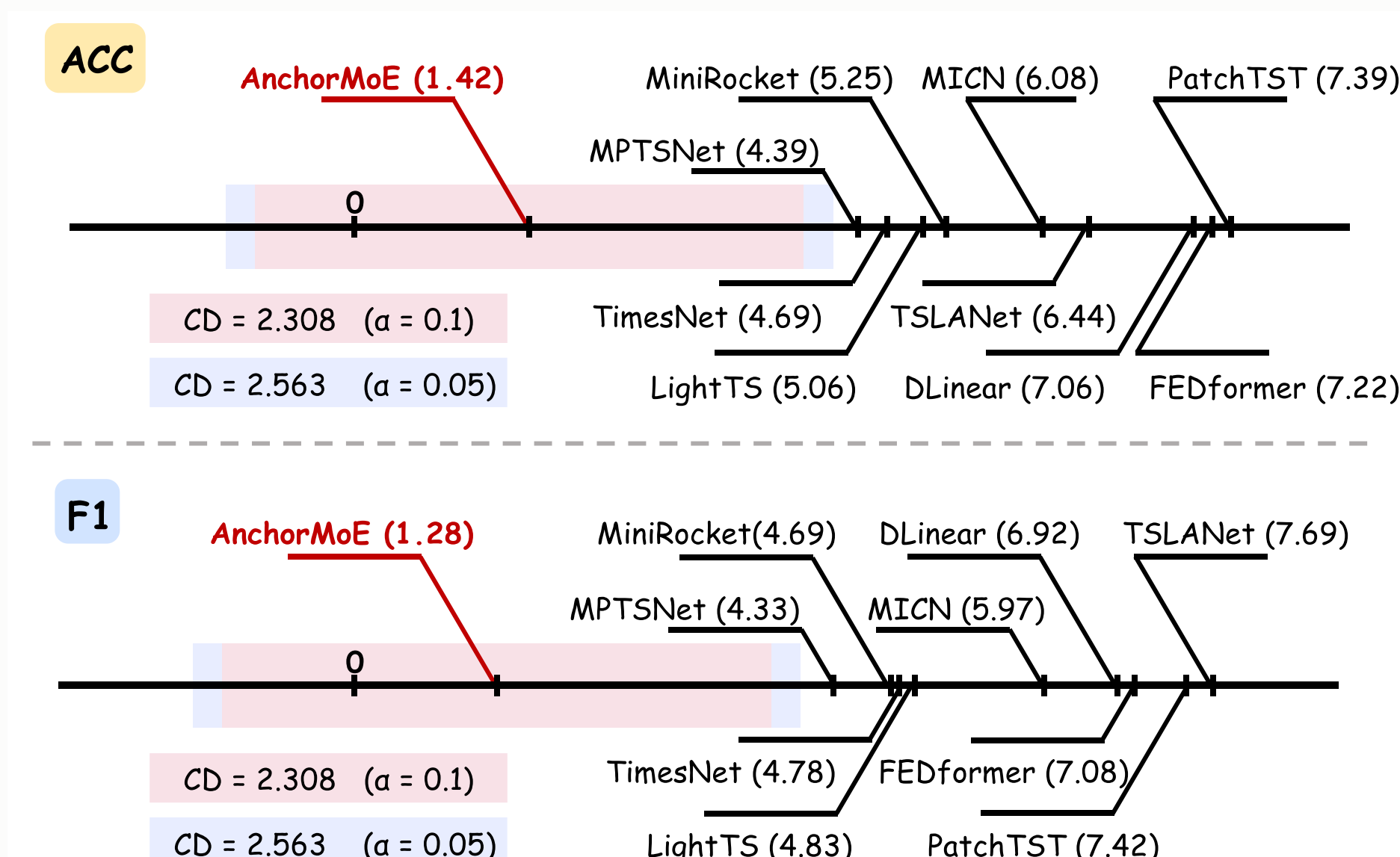


Figure 3. Statistical significance. AnchorMoE ranks first on ACC and F1; every baseline is beyond the Bonferroni-Dunn critical difference.

Locating the Key Patches

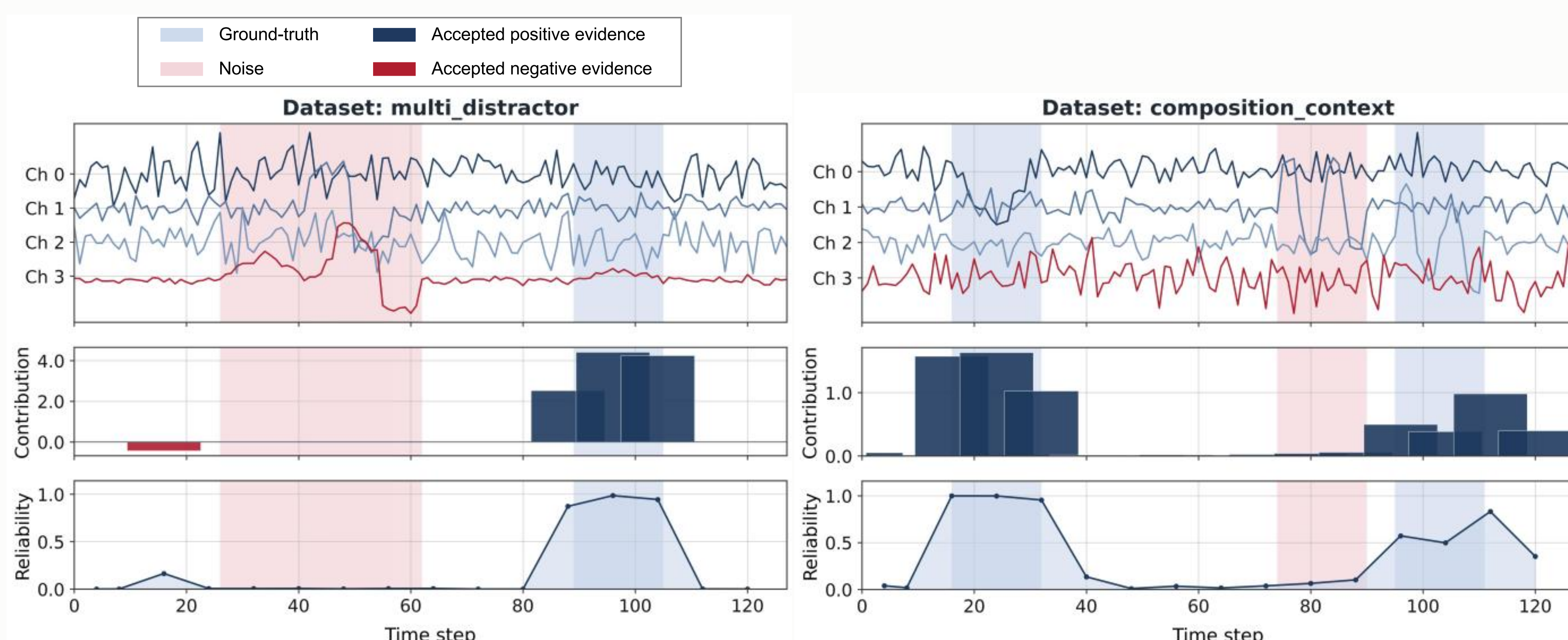


Figure 4. Key-patch localization. AnchorMoE highlights the annotated key regions while suppressing distractors and background.